

Record a game from the ball's view

When training for football games, coaches like to watch the game from several points-of-view in order to train their players better, develop strategies and improve the moves. Cameras and image processing software are very helpful in this task—providing close views from multiple angles, trajectory analysis and more. However, what has been missing till date is the view from the ball!

BallCam is a new technology developed by researchers at the Carnegie Mellon University and the University of Electro-Communications (UEC) in Tokyo, which allows a video camera to be embedded in a ball, thereby capturing smooth video as it spins through the air. While this might be illegal in official games, such recordings will surely be useful for training purposes. What is more, it might even vividly record the expression on a player's face when he misses a ball!

Inside: BallCam features a high-definition, wireless GoPro HERO 2 camcorder inserted into a hole on the side of a rubber-sheathed plastic foam football. (In the future, GoPro can be used with the company's yet-to-be-launched Wi-Fi control pack that will let the camera be controlled through smartphones.) One of the biggest problems in recording video from a spinning object as fast as a football is the resulting blur. BallCam uses a special algorithm to process the video from balls spinning at up to 600 rpm. The algorithm is designed to monitor and discard video frames that show only the sky. When the remaining frames are stitched together, the result is a relatively smooth, wide-angle view from the side of the football. The stitching software is similar to what NASA uses to stitch together images from its Mars rovers. So the technology is quite complex—and certainly not 'playful' stuff! The researchers say that the prototype can be enhanced to include multiple cameras and better camera sensors too.



Smart cane for elderly

At the Mobile World Congress 2013, Fujitsu demonstrated a New Generation Cane that can aid elders, specially-challenged, or even hikers in several ways. For starters, it can direct them properly to a programmed location by displaying arrows on the handle. It can monitor their heart rate, as well as environmental conditions. What is more, the associated Web application can be used to monitor the path that they take—to make sure they are always watched by their loved ones.

Inside: The smart cane is equipped with Bluetooth, GPS, Wi-Fi and cellular connectivity. The contoured handle is made of glossy plastic, and bulges on the top, where information is displayed using an array of LEDs. When none of the digital features is in use, all the lights disappear and the cane looks quite casual!

An accompanying app allows the caregiver to program some common locations using a PC or smartphone. Thereafter, when the user walks with the cane, it flashes a green arrow if the direction is right. If the user takes a wrong turn, it alerts using vibrations and red signs. It then guides the user back to the right direction using green arrows. GPS and cellular connectivity allows the user's walking path to be tracked using a Web application. A smart little step counter takes note of the number of times the stick is tapped! There is also a heart rate monitor built near the handle.

When the user presses a thumb on the sensor, it displays the heart rate on the handle, and also shares this information with caregivers. The cane also has built-in sensors that measure the temperature and humidity of the environment. This information is used to judge whether the user is comfortable. In case of any discomfort, the caregiver can remotely reprogram the cane to guide the user to a cooler or warmer place, or back to the home or hospital. The cane is battery-powered. At present, the prototype has a rather low battery life of 2-3 hours. This is likely to improve in the final product.

